

实验 2_1 嵌入式 Linux 环境的搭建

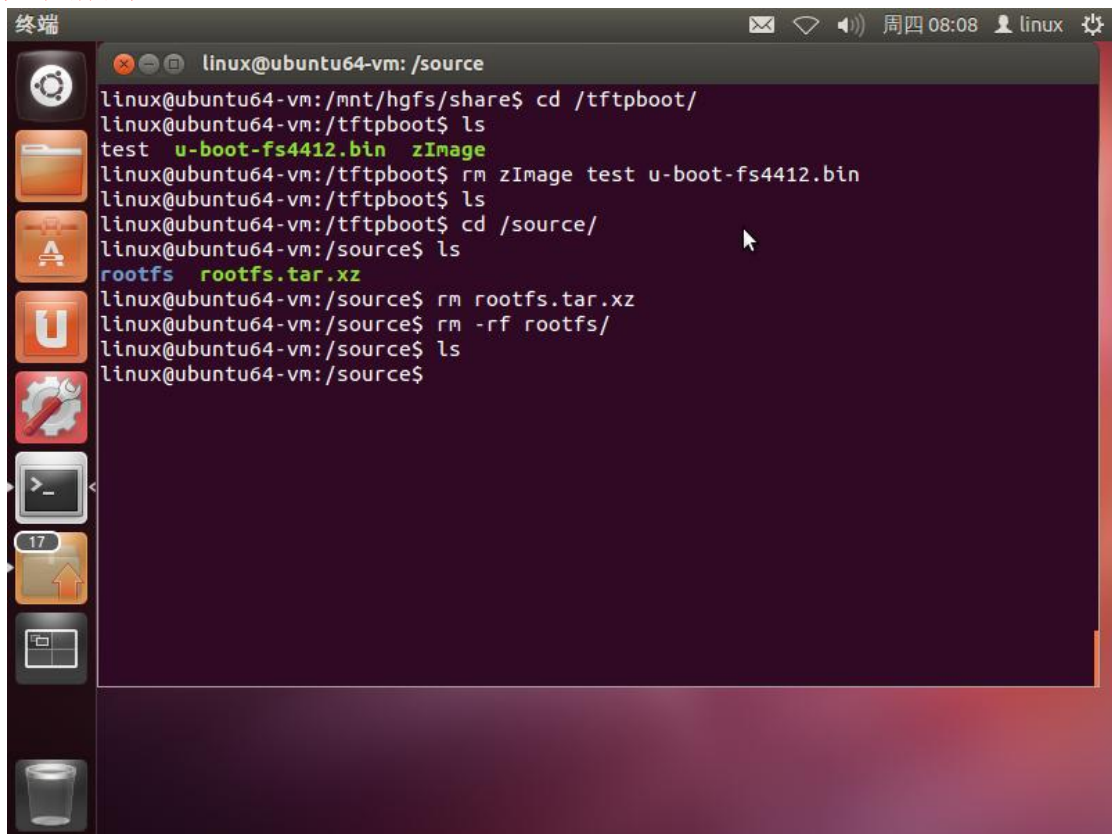
基本实验

约定：

1、在 `linux@ubuntu64-vm:~$` 下建立自己工作目录，在自己工作目录下建立三个文件夹 `app`、`kernel`、`drivers` (`mkdir -p app kernel drivers`)，今后凡是应用程序开发均在 `app` 下，凡是 Linux 系统（内核）开发均在 `kernel` 下、凡是驱动开发均在 `drivers` 下。

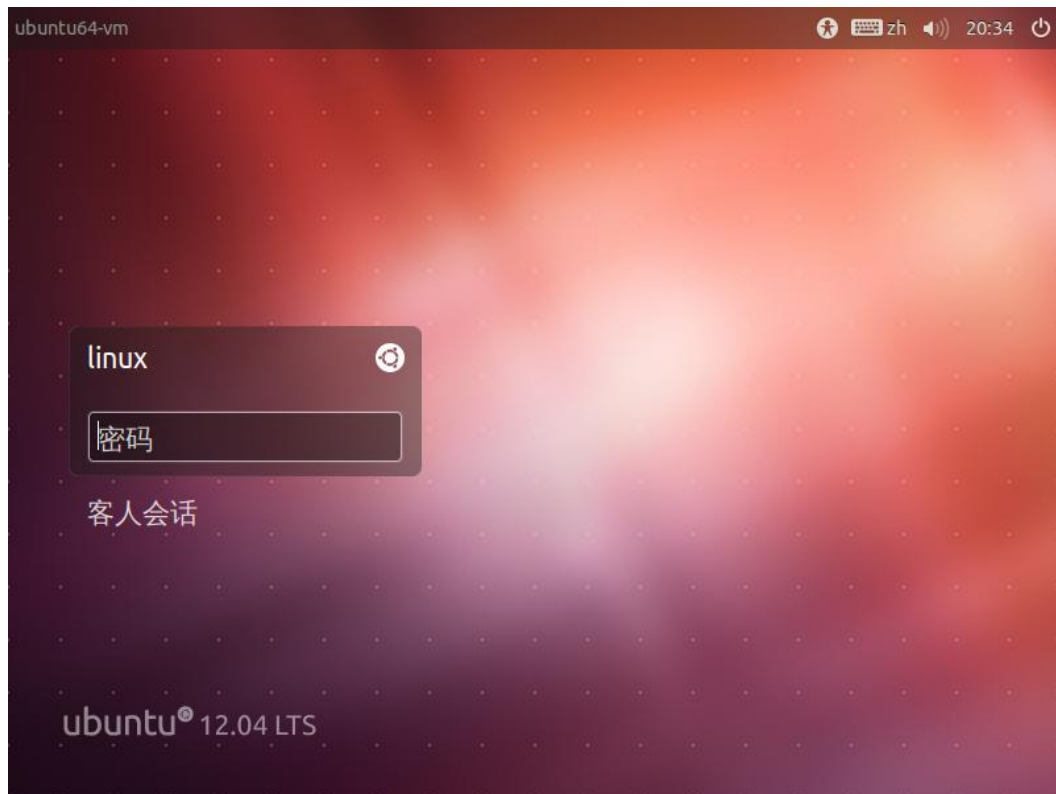
开发中严格区分开发机（pc 机上的 Linux，既宿主机、既 Ubuntu）和目标机（嵌入式 Linux，既实验箱、其系统信息显示在串口 `putty`、也可以在共享文件夹（`d:/share`）下建立建立三个文件夹 `app`、`kernel`、`drivers` 对应）

2 实验用 `/tftp` 和 `/source/rootfs/` 目录如果已经有内容,可以删除,操作如图所示

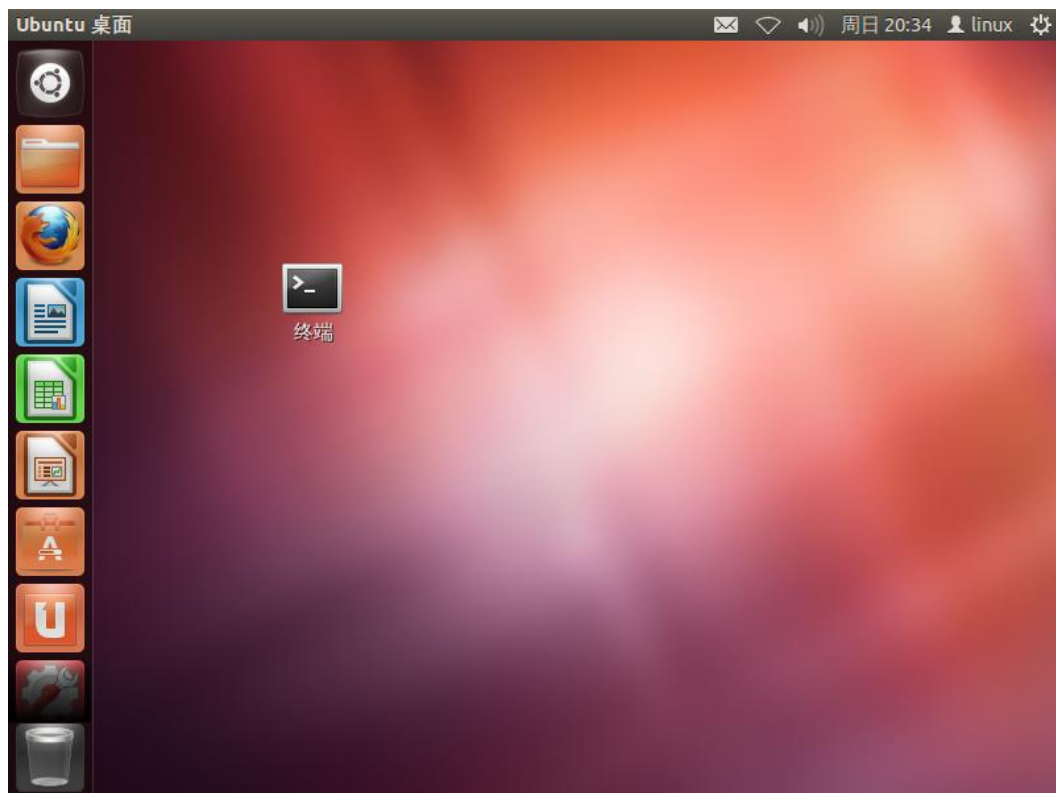


```
终端
linux@ubuntu64-vm: /source
linux@ubuntu64-vm:/mnt/hgfs/share$ cd /tftpboot/
linux@ubuntu64-vm:/tftpboot$ ls
test  u-boot-fs4412.bin  zImage
linux@ubuntu64-vm:/tftpboot$ rm zImage test u-boot-fs4412.bin
linux@ubuntu64-vm:/tftpboot$ ls
linux@ubuntu64-vm:/tftpboot$ cd /source/
linux@ubuntu64-vm:/source$ ls
rootfs  rootfs.tar.xz
linux@ubuntu64-vm:/source$ rm rootfs.tar.xz
linux@ubuntu64-vm:/source$ rm -rf rootfs/
linux@ubuntu64-vm:/source$ ls
linux@ubuntu64-vm:/source$
```

第一步：启动虚拟机（已安装好）。输入密码：**1**（sudo 密码：**1**）

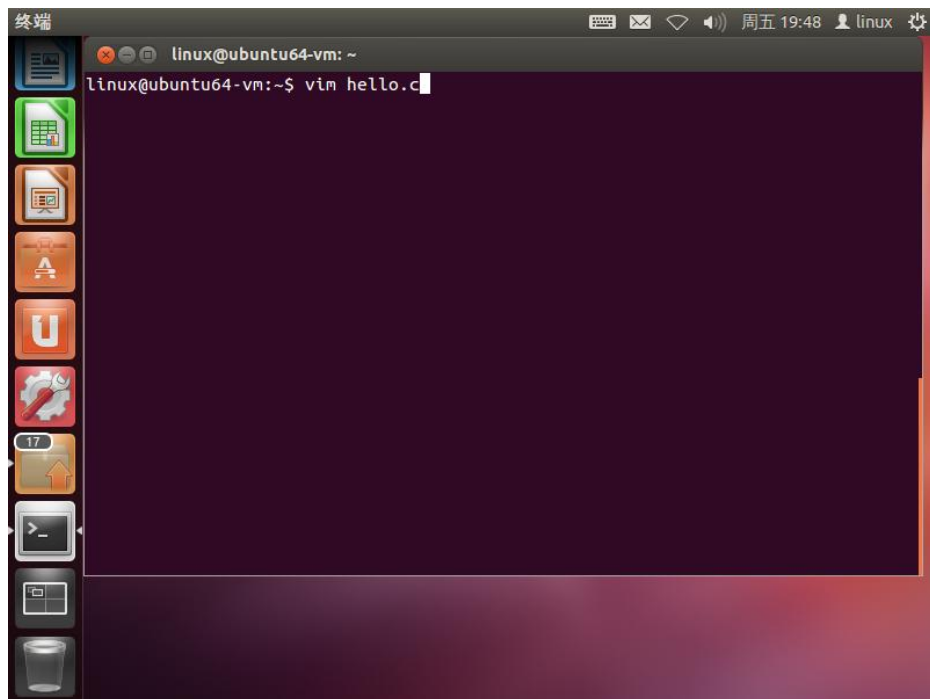


登录后的桌面如下：

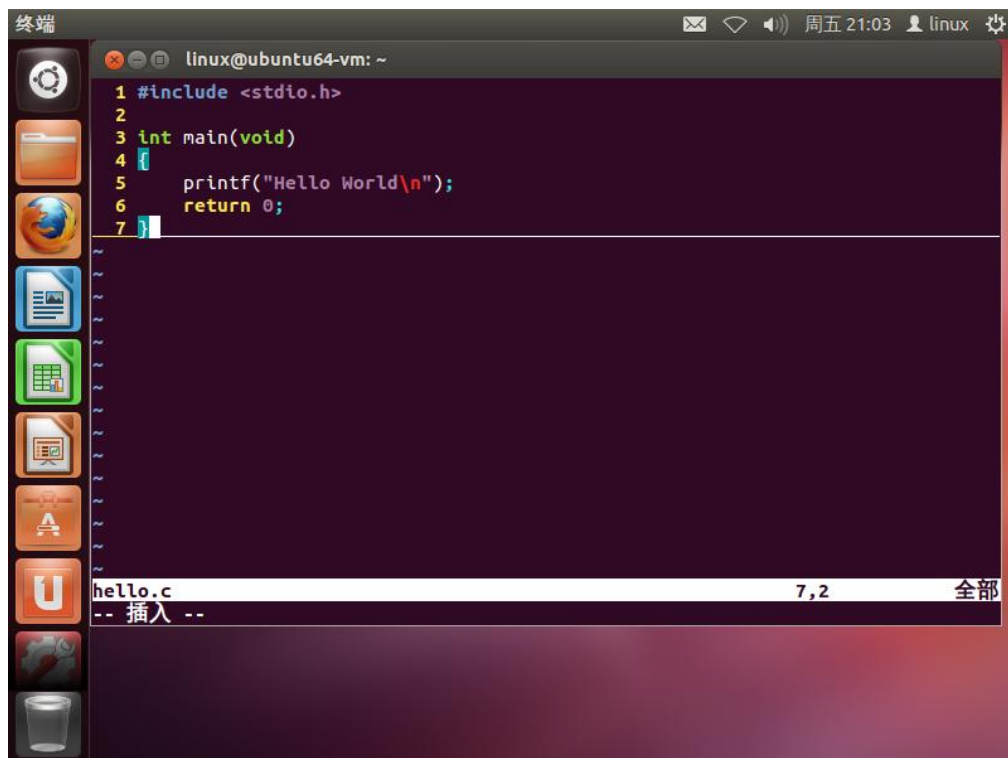


第二步：点击“终端”，输入命令

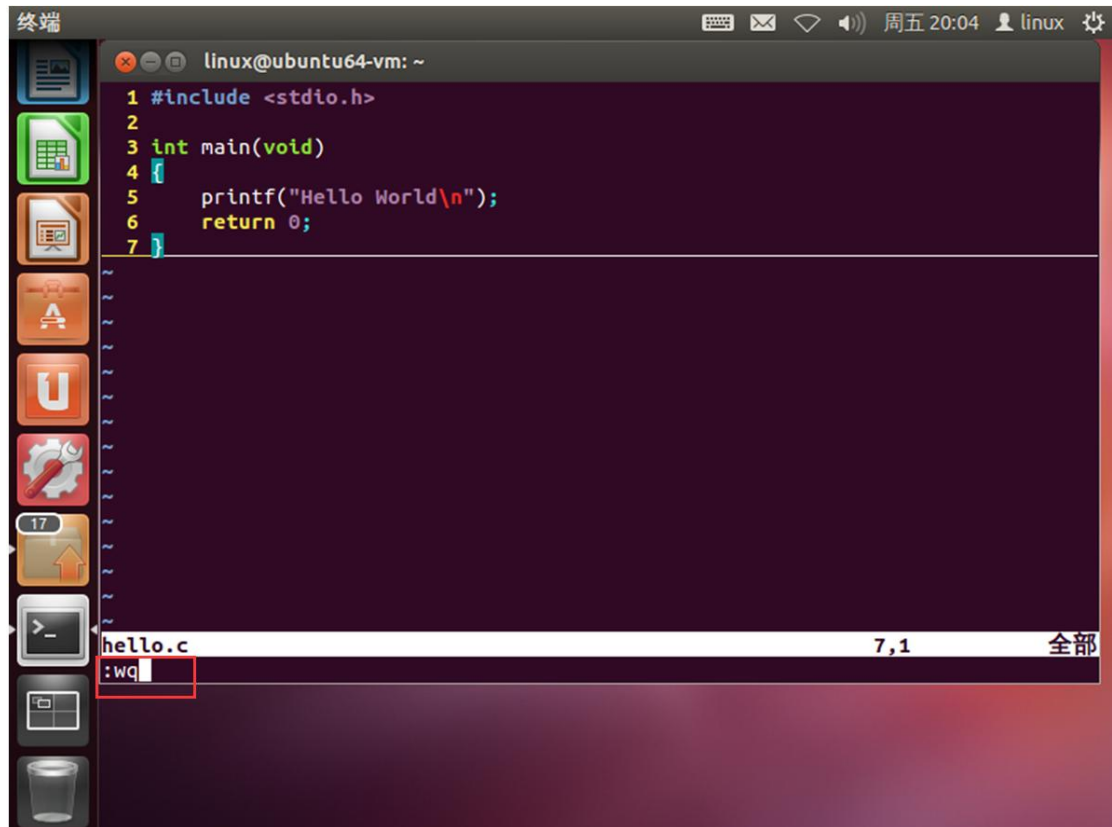
`$ vim hello.c`（在 **app** 下使用 **vim** 创建或者打开 **hello.c**）



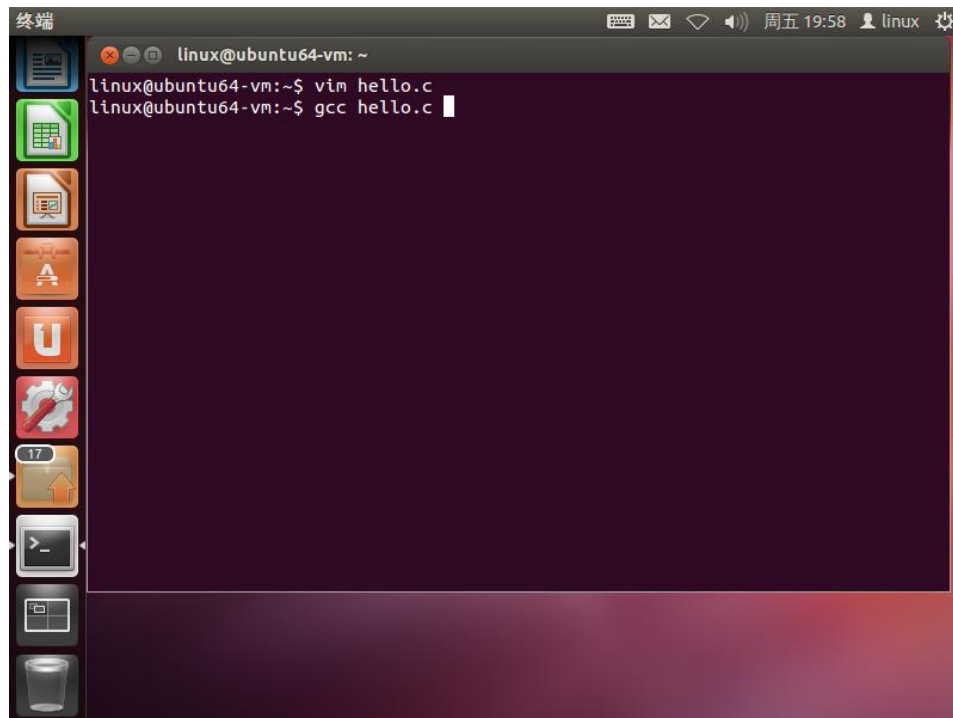
第三步：进入 vim 后敲击键盘上的 a 按键，进入编辑模式，编辑“Hello Wrold”程序。



:wq



第五步：回到终端后，使用 gcc 编译 hello.c，输入命令：\$ gcc hello.c

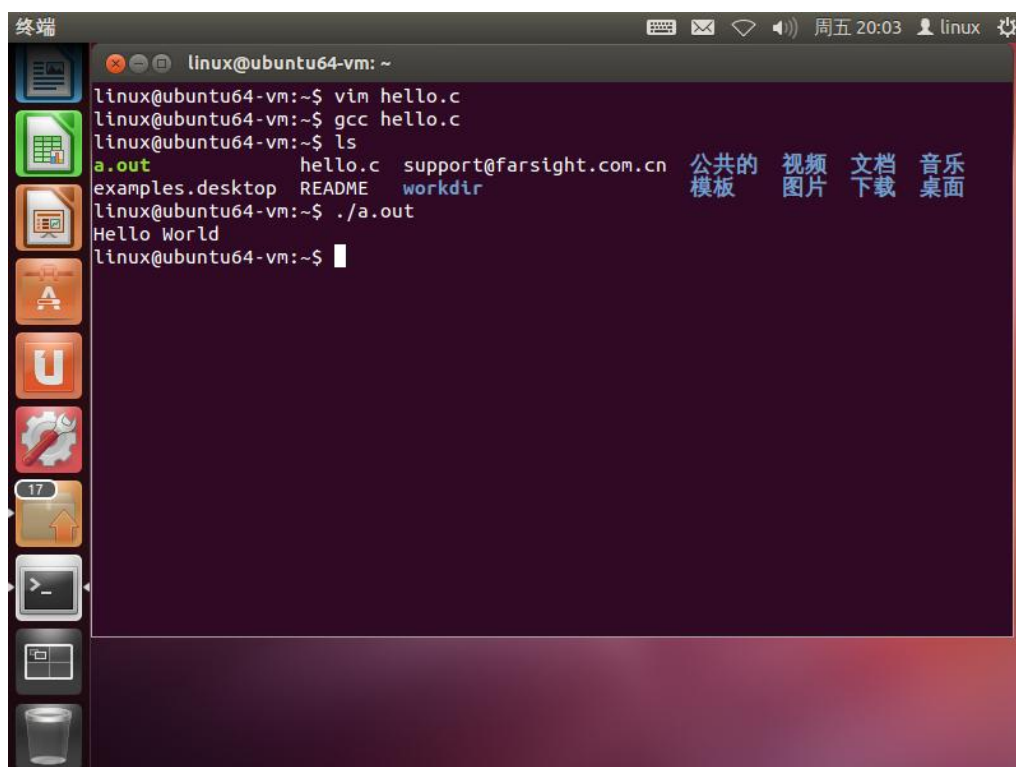


A terminal window titled '终端' (Terminal) with a dark purple background. The prompt is 'linux@ubuntu64-vm: ~'. The user has entered 'vim hello.c' and then 'gcc hello.c'. The window has a sidebar on the left with various application icons and a top bar with system status icons and the time '周五 19:58'.

```
linux@ubuntu64-vm: ~
linux@ubuntu64-vm:~$ vim hello.c
linux@ubuntu64-vm:~$ gcc hello.c
```

第六步：编译成功后，使用 ls 命令查看编译后程序 a.out，并且执行 a.out —— 在终端打印 Hello World

\$./a.out



A terminal window titled '终端' (Terminal) with a dark purple background. The prompt is 'linux@ubuntu64-vm: ~'. The user has entered a series of commands: 'vim hello.c', 'gcc hello.c', 'ls', and './a.out'. The output of 'ls' shows 'a.out', 'hello.c', 'support@farsight.com.cn', 'examples.desktop', 'README', and 'workdir'. The output of './a.out' is 'Hello World'. The window has a sidebar on the left with various application icons and a top bar with system status icons and the time '周五 20:03'.

```
linux@ubuntu64-vm: ~
linux@ubuntu64-vm:~$ vim hello.c
linux@ubuntu64-vm:~$ gcc hello.c
linux@ubuntu64-vm:~$ ls
a.out      hello.c    support@farsight.com.cn  公共的  视频  文档  音乐
examples.desktop  README    workdir                  模板   图片  下载  桌面
linux@ubuntu64-vm:~$ ./a.out
Hello World
linux@ubuntu64-vm:~$
```

第七步：查看有关环境配置的正确性

使用 ifconfig 命令查看 ip 正确性

```
linux@ubuntu64-vm: ~  
linux@ubuntu64-vm:~$ ifconfig  
eth0      Link encap:以太网  硬件地址 00:0c:29:87:2d:9c  
          inet 地址 192.168.100.192 广播:192.168.1.255 掩码:255.255.255.0  
          inet6 地址: fe80::20c:29ff:fe87:2d9c/64 Scope:Link  
          UP BROADCAST RUNNING MULTICAST  MTU:1500  跃点数:1  
          接收数据包:97 错误:0 丢弃:0 过载:0 帧数:0  
          发送数据包:92 错误:0 丢弃:0 过载:0 载波:0  
          碰撞:0 发送队列长度:1000  
          接收字节:19160 (19.1 KB) 发送字节:10348 (10.3 KB)  
  
lo        Link encap:本地环回  
          inet 地址:127.0.0.1 掩码:255.0.0.0  
          inet6 地址: ::1/128 Scope:Host  
          UP LOOPBACK RUNNING  MTU:16436 跃点数:1  
          接收数据包:33 错误:0 丢弃:0 过载:0 帧数:0  
          发送数据包:33 错误:0 丢弃:0 过载:0 载波:0  
          碰撞:0 发送队列长度:0  
          接收字节:2866 (2.8 KB) 发送字节:2866 (2.8 KB)  
  
linux@ubuntu64-vm:~$
```

第八步：工具链的测试（查看交叉开发工具链配置正确）

\$ arm-none-linux-gnueabi-gcc -v

如出现以下错误：

```
linux@ubuntu64-vm: ~  
linux@ubuntu64-vm:~$ arm-none-linux-gnueabi-gcc -v  
arm-none-linux-gnueabi-gcc: error: -v: No such file or directory  
arm-none-linux-gnueabi-gcc: fatal error: no input files  
compilation terminated.  
linux@ubuntu64-vm:~$  
  
linux@ubuntu64-vm:~$ uname -a  
Linux ubuntu64-vm 3.2.0-24-generic #37-Ubuntu SMP Wed Apr 25 08:43:22 UTC 2012 x86_64 x86_64 x86_64 GNU/Linux  
linux@ubuntu64-vm:~$
```

直接输入命令，即可执行，这样说明我们的交叉工具链就安装好了

二、交叉开发过程指导

1、在 Windows 下，复制“资源\rootfs.tar.xz”，“资源\zImage”复制到所创建的共享目录下（例如 D:\share）



2、将共享目录中需要下载的文件复制到 tftp 目录中
在 Ubuntu 下，将 zImage 文件复制到/tftpboot 目录下。

`$cp /mnt/hgfs/share/zImage /tftpboot/`

```
linux@ubuntu64-vm:~$ cp /mnt/hgfs/share/zImage /tftpboot/
linux@ubuntu64-vm:~$ ls /tftpboot/
test  zImage
linux@ubuntu64-vm:~$
```

3、解压文件系统

复制 rootfs.tar.bz2 文件到虚拟机 Ubuntu 下的 /source 目录下

```
$ cp /mnt/hgfs/share/rootfs.tar.xz /source/
```

```
linux@ubuntu64-vm:~$ cp /mnt/hgfs/share/rootfs.tar.xz /source/
linux@ubuntu64-vm:~$ ls /source/
rootfs.tar.xz
linux@ubuntu64-vm:~$
```

```
$ cd /source
```

```
$ tar xvf rootfs.tar.xz
```

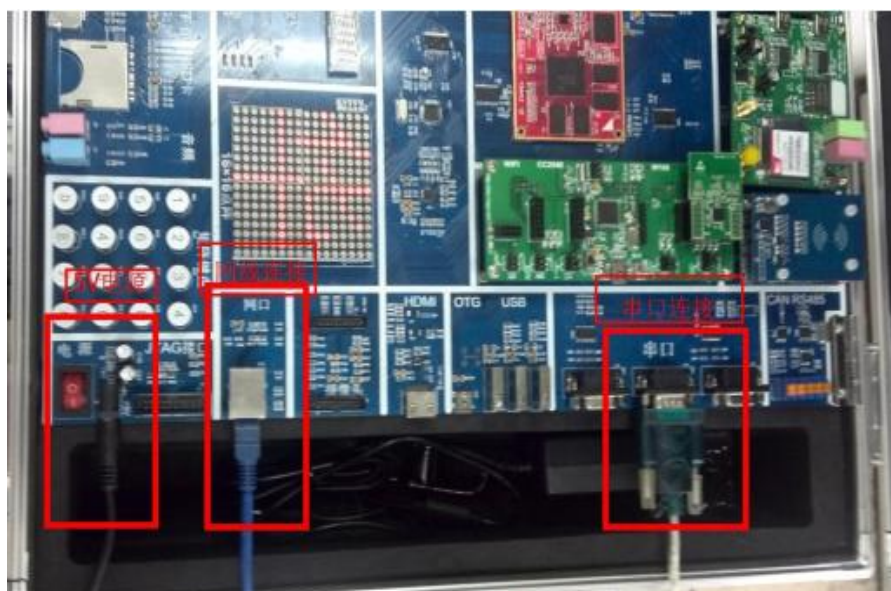
```
linux@ubuntu64-vm:~$ cd /source/
linux@ubuntu64-vm:/source$ ls
rootfs.tar.xz
linux@ubuntu64-vm:/source$ tar xvf rootfs.tar.xz
```

执行解压后的结果

```
rootfs/bin/ps
rootfs/bin/busybox
rootfs/bin/kill
rootfs/bin/cp
rootfs/bin/gunzip
rootfs/bin/df
rootfs/bin/sleep
rootfs/bin/pwd
rootfs/bin/adduser
rootfs/bin/ttyhack
rootfs/bin/getopt
rootfs/bin/sed
rootfs/bin/delgroup
rootfs/bin/tonice
rootfs/bin/chgrp
rootfs/bin/chown
rootfs/bin/watch
rootfs/bin/ping6
rootfs/bin/sync
rootfs/bin/mt
rootfs/bin/run-parts
rootfs/bin/rpm
rootfs/bin/su
rootfs/bin/mount
rootfs/bin/catv
rootfs/bin/tar
rootfs/proc/
rootfs/sys/
linux@ubuntu64-vn:/source$
```

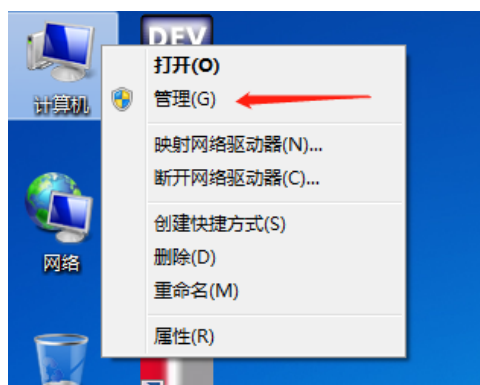
4、连接开发板

将串口线一端与实验箱的串口 2 连接，另一端连接到 PC 的 USB 接口。将网线一端与实验箱的网口连接，另一端连接到 PC 的网口。

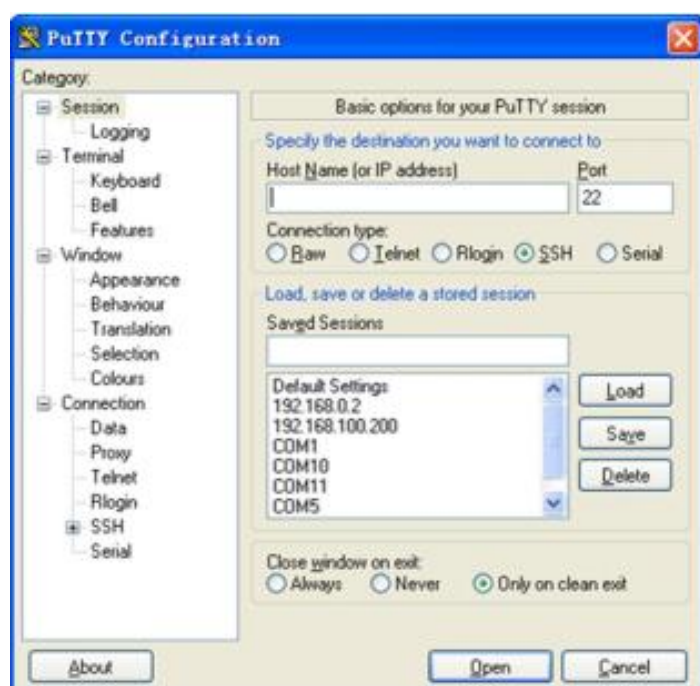
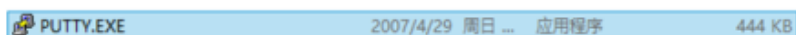


5、设置串口调试工具

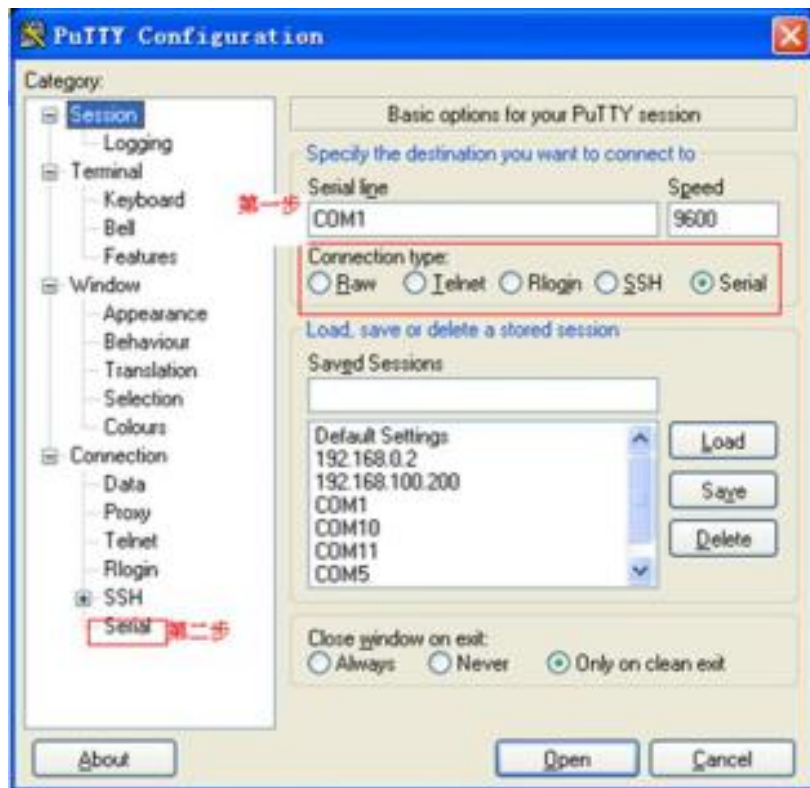
关闭实验箱电源，使用串口线将试验箱与 PC 连接。查看计算机设备管理器，找到 CH340 对应的端口号（例如 COM3）



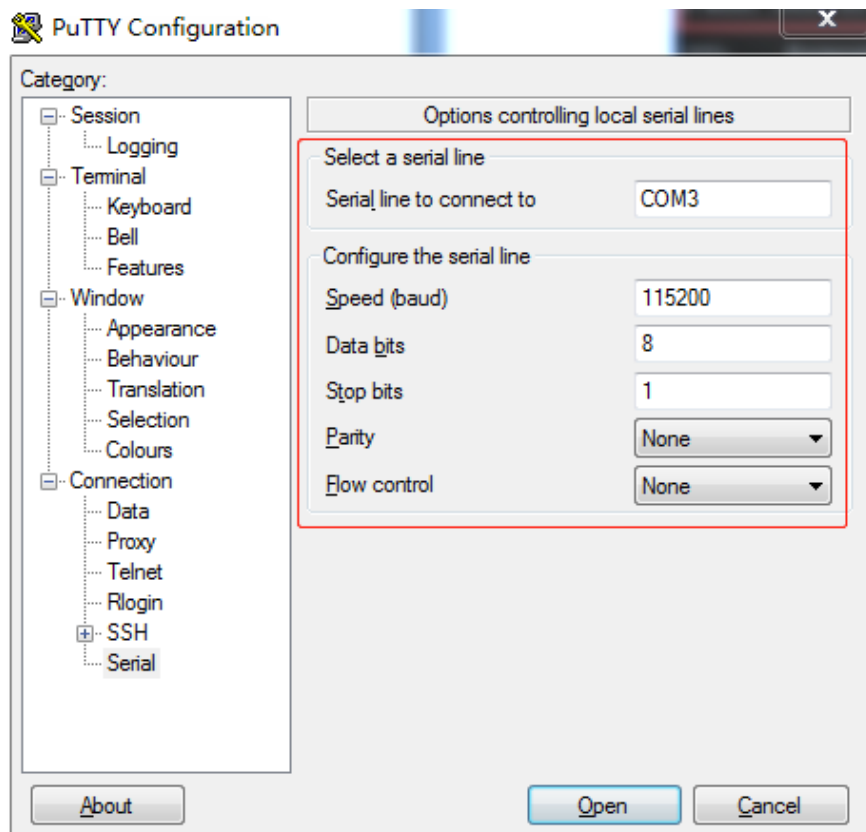
打开 Putty，通过串口把开发板的调试数据显示在 PC 上。



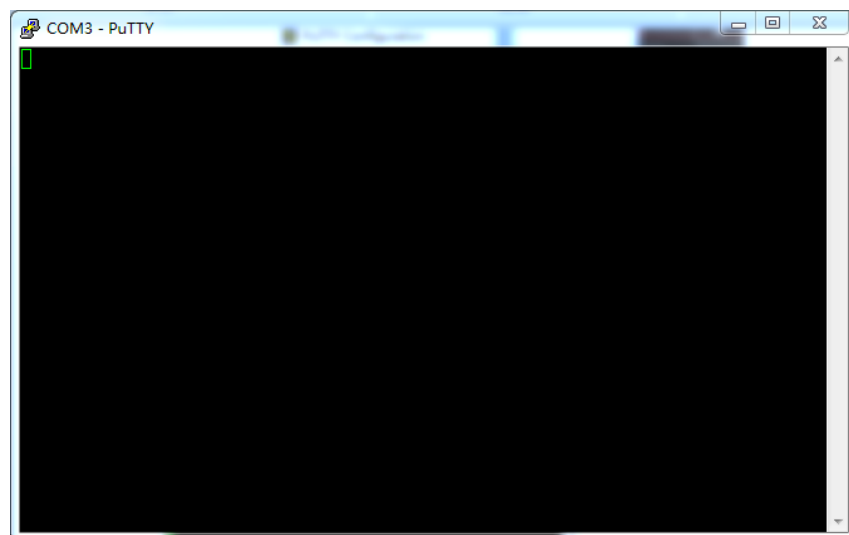
选中第一步方框内的 Serial,再点击第二步中的 Serial, 进入串口设置的对话框



填写与 CH340 对应的端口号 (COM3), 其他选项根据下图填写进行配置



点击 open 打开串口



6、启动开发板

拨码开关拨至 0110，如下图所示



启动开发板，在倒计时结束前（5s），按任意键停止在 Uboot 处，串口终端显示图如下图所示

```
COM3 - PuTTY
U-Boot 2010.03 (Sep 10 2014 - 12:01:58) For FarSight FS4412 eMMC

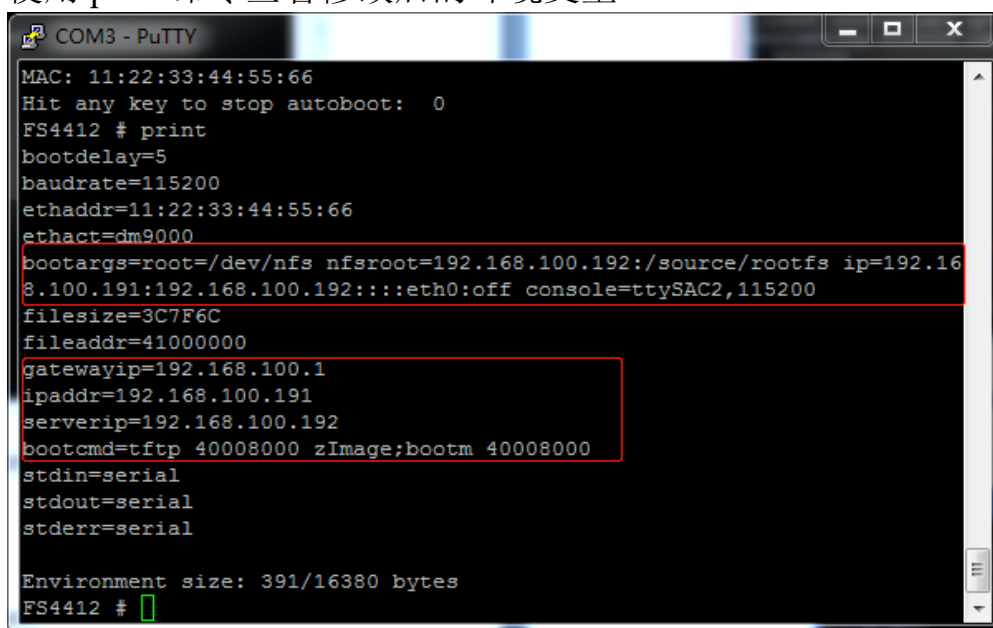
APLL = 1000MHz, MPLL = 800MHz
ARM_CLOCK = 1000MHz
PMIC: Pls check the i2c @ pmic, id = 21,error
Board: FS4412
DRAM: 2 GB
MMC0: 14910 MB
MMC1: 7701 MB
In: serial
Out: serial
Err: serial

Checking Boot Mode ... eMMC
Net: dm9000
dm9000 i/o: 0x50000000, id: 0x90000a46
DM9000: running in 16 bit mode
MAC: 11:22:33:44:55:66
Hit any key to stop autoboot: 0
FS4412 #
```

7、输入如下命令修改开发板环境变量（在两个#之间的内容为一条命令）

```
#setenv serverip 192.168.100.192
#setenv ipaddr 192.168.100.191
#setenv gatewayip 192.168.100.1
#setenv bootcmd tftp 40008000 zImage\; bootm 40008000
#setenv bootargs root=/dev/nfs nfsroot=192.168.100.192:/source/rootfs
ip=192.168.100.191:192.168.100.192:::eth0:off
console=ttySAC2,115200
#saveenv
```

8、使用 print 命令查看修改后的环境变量。



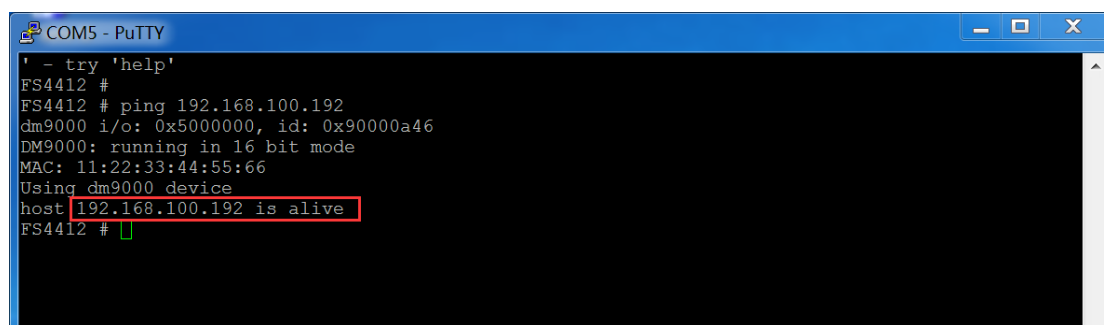
```
COM3 - PuTTY
MAC: 11:22:33:44:55:66
Hit any key to stop autoboot: 0
FS4412 # print
bootdelay=5
baudrate=115200
ethaddr=11:22:33:44:55:66
ethact=dm9000
bootargs=root=/dev/nfs nfsroot=192.168.100.192:/source/rootfs ip=192.168.100.191:192.168.100.192:::eth0:off console=ttySAC2,115200
filesize=3C7F6C
fileaddr=41000000
gatewayip=192.168.100.1
ipaddr=192.168.100.191
serverip=192.168.100.192
bootcmd=tftp 40008000 zImage;bootm 40008000
stdin=serial
stdout=serial
stderr=serial

Environment size: 391/16380 bytes
FS4412 #
```

9、测试开发板与虚拟机网络是否连通

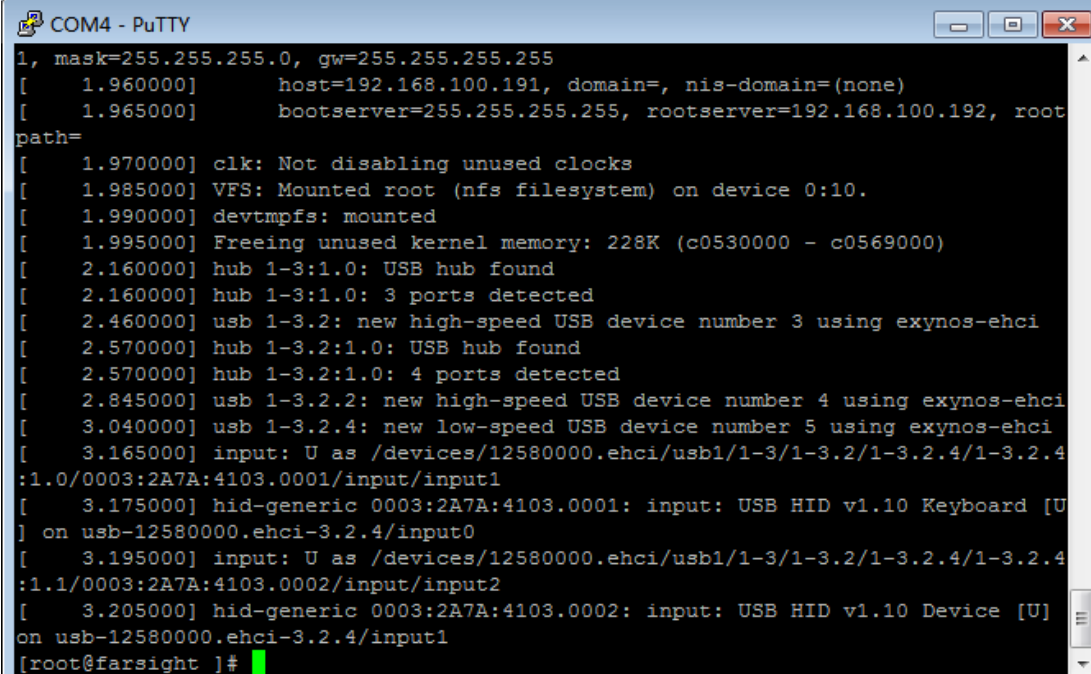
使用 ping 命令尝试 ping 一下 Ubuntu 主机，如下图所示，表示网络已经联通

```
# ping 192.168.100.192
```



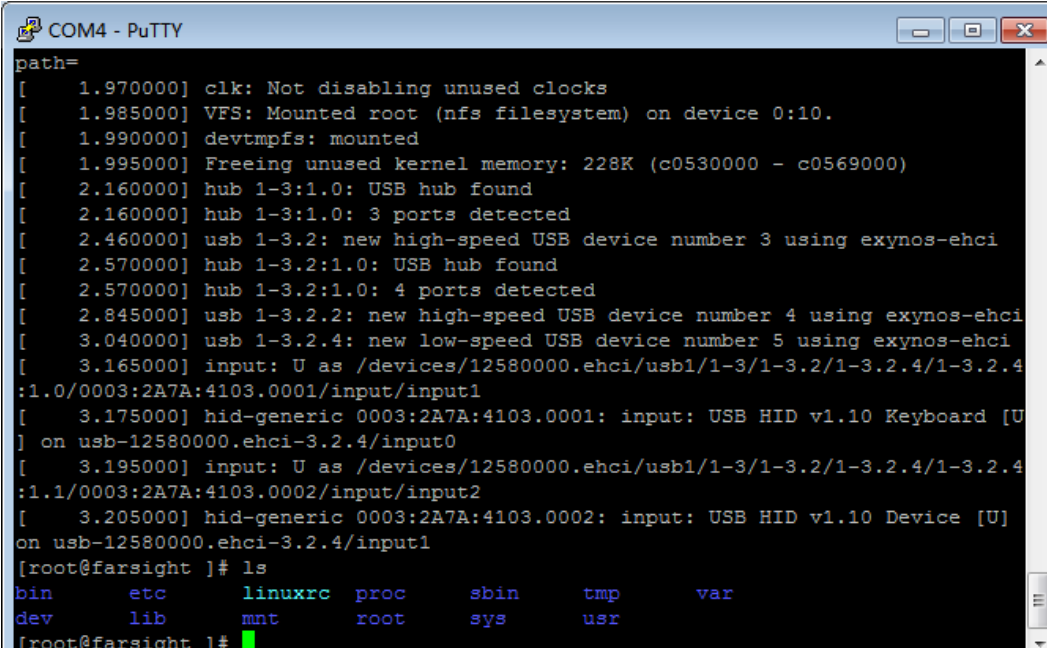
```
COM5 - PuTTY
' - try 'help'
FS4412 #
FS4412 # ping 192.168.100.192
dm9000 i/o: 0x5000000, id: 0x90000a46
DM9000: running in 16 bit mode
MAC: 11:22:33:44:55:66
Using dm9000 device
host 192.168.100.192 is alive
FS4412 #
```

10、重启开发板，等待倒计时，如下图启动，表明内核从主机的 /tftpboot 处下载，文件系统为 nfs 网络文件系统，位置为主机的 /source/rootfs/ 处



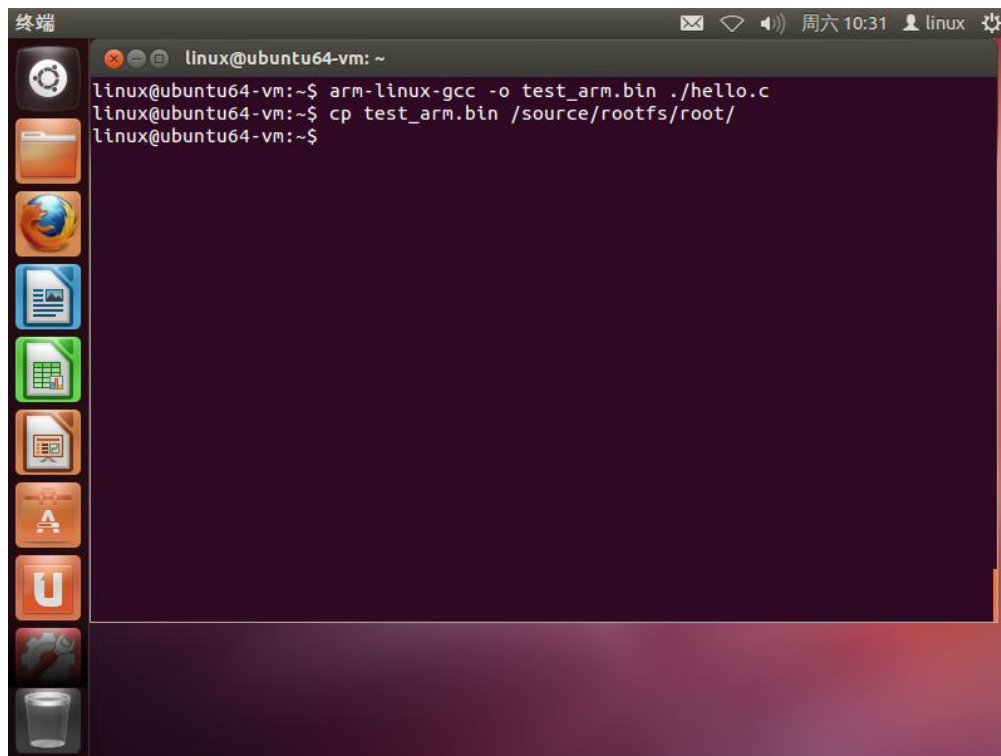
```
COM4 - PuTTY
1, mask=255.255.255.0, gw=255.255.255.255
[ 1.960000] host=192.168.100.191, domain=, nis-domain=(none)
[ 1.965000] bootserver=255.255.255.255, rootserver=192.168.100.192, root
path=
[ 1.970000] clk: Not disabling unused clocks
[ 1.985000] VFS: Mounted root (nfs filesystem) on device 0:10.
[ 1.990000] devtmpfs: mounted
[ 1.995000] Freeing unused kernel memory: 228K (c0530000 - c0569000)
[ 2.160000] hub 1-3:1.0: USB hub found
[ 2.160000] hub 1-3:1.0: 3 ports detected
[ 2.460000] usb 1-3.2: new high-speed USB device number 3 using exynos-ehci
[ 2.570000] hub 1-3.2:1.0: USB hub found
[ 2.570000] hub 1-3.2:1.0: 4 ports detected
[ 2.845000] usb 1-3.2.2: new high-speed USB device number 4 using exynos-ehci
[ 3.040000] usb 1-3.2.4: new low-speed USB device number 5 using exynos-ehci
[ 3.165000] input: U as /devices/12580000.ehci/usb1/1-3/1-3.2/1-3.2.4/1-3.2.4
:1.0/0003:2A7A:4103.0001/input/input1
[ 3.175000] hid-generic 0003:2A7A:4103.0001: input: USB HID v1.10 Keyboard [U
] on usb-12580000.ehci-3.2.4/input0
[ 3.195000] input: U as /devices/12580000.ehci/usb1/1-3/1-3.2/1-3.2.4/1-3.2.4
:1.1/0003:2A7A:4103.0002/input/input2
[ 3.205000] hid-generic 0003:2A7A:4103.0002: input: USB HID v1.10 Device [U
] on usb-12580000.ehci-3.2.4/input1
[root@farsight]#
```

11、我们 **ls** 一下，此时显示的文件系统是通过 nfs 挂载了虚拟机 /source/rootfs 的文件系统



```
COM4 - PuTTY
path=
[ 1.970000] clk: Not disabling unused clocks
[ 1.985000] VFS: Mounted root (nfs filesystem) on device 0:10.
[ 1.990000] devtmpfs: mounted
[ 1.995000] Freeing unused kernel memory: 228K (c0530000 - c0569000)
[ 2.160000] hub 1-3:1.0: USB hub found
[ 2.160000] hub 1-3:1.0: 3 ports detected
[ 2.460000] usb 1-3.2: new high-speed USB device number 3 using exynos-ehci
[ 2.570000] hub 1-3.2:1.0: USB hub found
[ 2.570000] hub 1-3.2:1.0: 4 ports detected
[ 2.845000] usb 1-3.2.2: new high-speed USB device number 4 using exynos-ehci
[ 3.040000] usb 1-3.2.4: new low-speed USB device number 5 using exynos-ehci
[ 3.165000] input: U as /devices/12580000.ehci/usb1/1-3/1-3.2/1-3.2.4/1-3.2.4
:1.0/0003:2A7A:4103.0001/input/input1
[ 3.175000] hid-generic 0003:2A7A:4103.0001: input: USB HID v1.10 Keyboard [U
] on usb-12580000.ehci-3.2.4/input0
[ 3.195000] input: U as /devices/12580000.ehci/usb1/1-3/1-3.2/1-3.2.4/1-3.2.4
:1.1/0003:2A7A:4103.0002/input/input2
[ 3.205000] hid-generic 0003:2A7A:4103.0002: input: USB HID v1.10 Device [U
] on usb-12580000.ehci-3.2.4/input1
[root@farsight]# ls
bin      etc      linuxrc  proc     sbin     tmp      var
dev      lib      mnt      root     sys      usr
[root@farsight]#
```

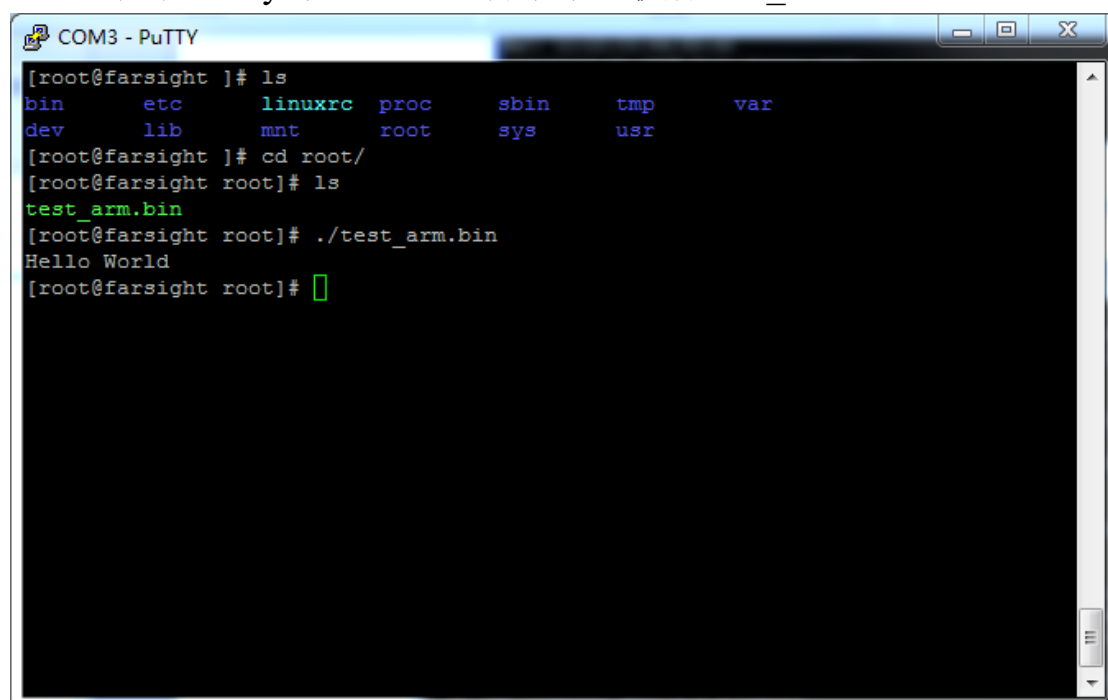
12、在 Ubuntu 下，使用配置好交叉编译工具编译编译所编写的“Hello World”程序(如图)，将所得二进制文件命名为 test_arm.bin，复制到 /source/rootfs/root 中

A terminal window titled '终端' (Terminal) with a dark background and light text. The window shows the following commands and output:

```
linux@ubuntu64-vm: ~  
linux@ubuntu64-vm:~$ arm-linux-gcc -o test_arm.bin ./hello.c  
linux@ubuntu64-vm:~$ cp test_arm.bin /source/rootfs/root/  
linux@ubuntu64-vm:~$
```

The window has a sidebar on the left with various application icons and a top bar with system status icons and the date '周六 10:31'.

13、然后在 Putty 下进入 root 目录下，执行 test_arm.bin

A Putty terminal window titled 'COM3 - PuTTY' with a black background and white text. The window shows the following commands and output:

```
[root@farsight ~]# ls  
bin      etc      linuxrc  proc     sbin     tmp      var  
dev      lib      mnt      root     sys      usr  
[root@farsight ~]# cd root/  
[root@farsight root]# ls  
test_arm.bin  
[root@farsight root]# ./test_arm.bin  
Hello World  
[root@farsight root]#
```

The window has a standard window title bar and a scrollbar on the right side.

至此，我们可以通过此方式对开发板进行调试了!!!

实验 Ubuntu、嵌入式 Linux 下的菜单 C 语言程序开发(要检查)

(以下标黄的每一步的结果, 都需要截图保存, 在雨课堂提交为: 20231000000+姓名+实验 2-1.pdf)

一、cp led_testmenu.c 到 app 文件夹下,

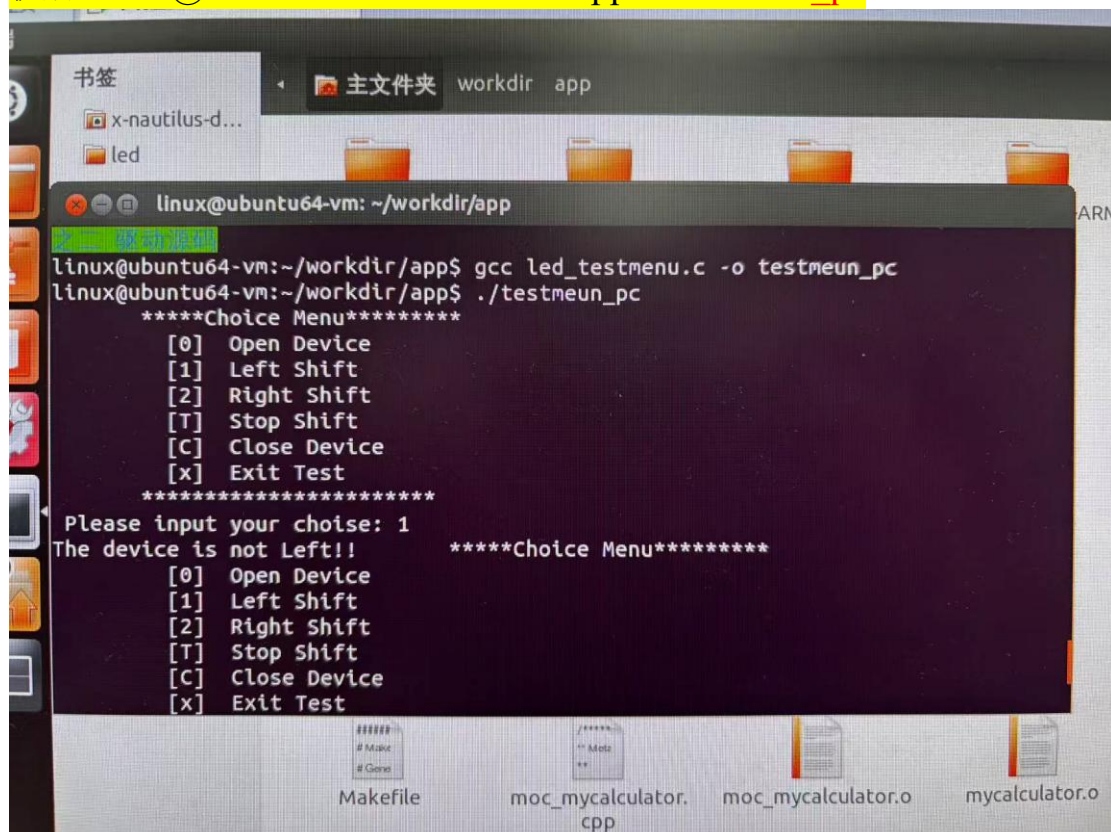
```
linux@ubuntu64-vm:~/workdir/app$ cp  
/mnt/hgfs/share/app/led_testmenu.c ./
```

二、Ubuntu 下编译执行 C 程序

编译 c 程序: led_testmenu.c

```
编译 nux@ubuntu64-vm:~/workdir/app$ gcc led_testmenu.c -o  
testmeun_pc
```

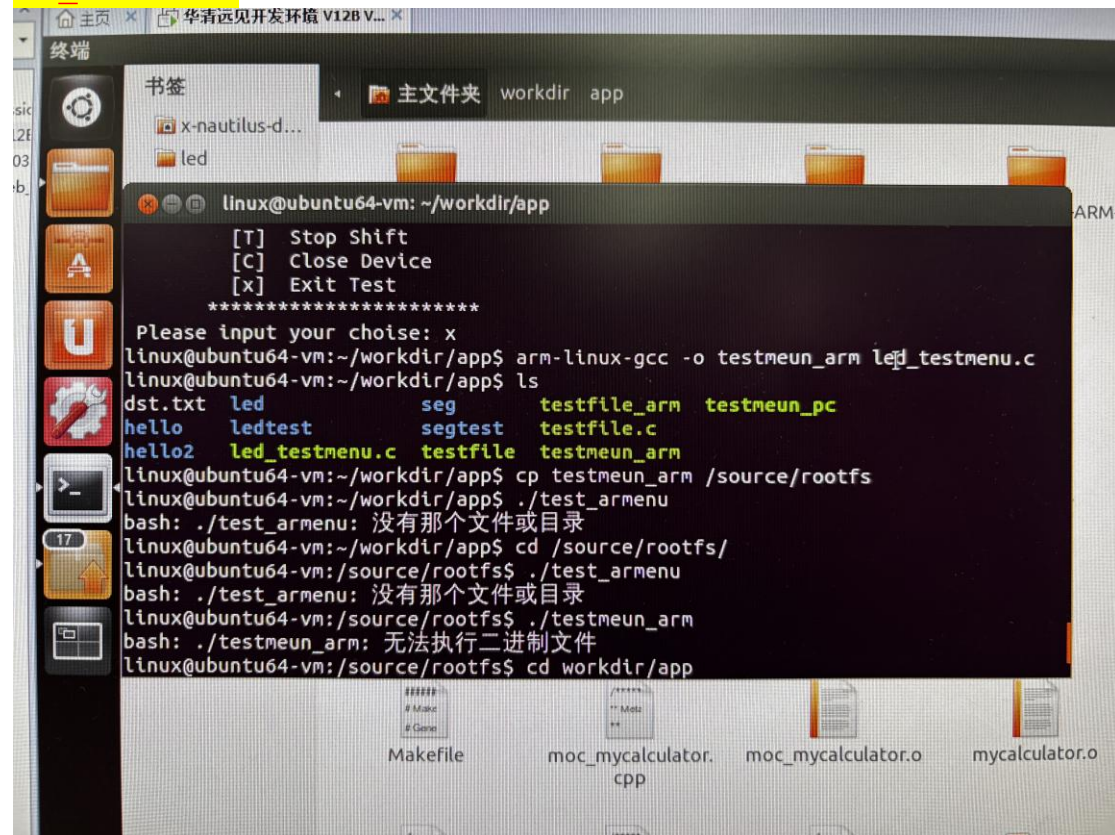
```
执行 linux@ubuntu64-vm:~/workdir/app$ ./testmeun_pc
```



三 Ubuntu 下编译 C 程序,嵌入式 linux 系统下执行 C 程序

1 编译 c 程序: led_testmenu.c

```
linux@ubuntu64-vm:~/workdir/app$ arm-linux-gcc -o testmeun_arm  
led_testmenu.c
```



2 linux@ubuntu64-vm:~/workdir/app\$ ls

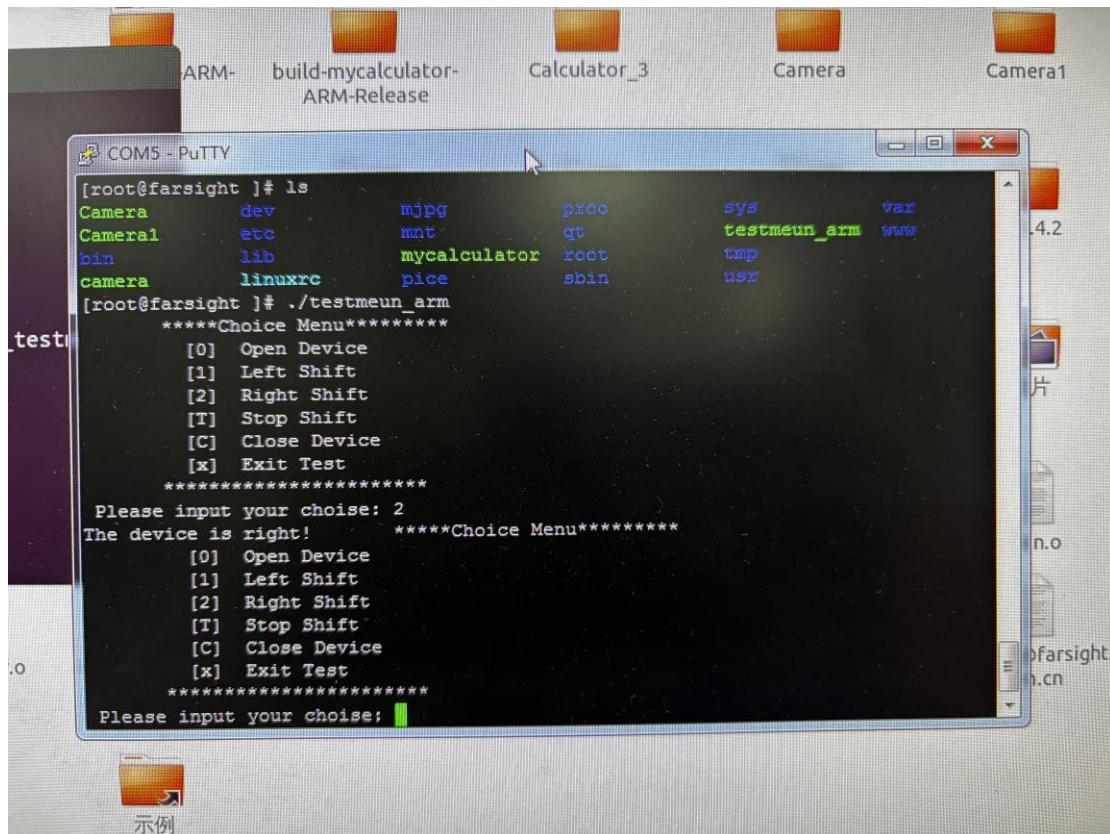
3、拷贝到挂载的文件系统:

```
linux@ubuntu64-vm:~/workdir/app$ cp testmeun_arm /source/rootfs
```

4、打开串口终端 (putty)

5、启动实验箱嵌入式 Linux

6、执行 ./test_armenu



根据菜单操作，并修改（合适的设计、修改可以有创新分）

（以下完成的话，可以有创新分，执行过程和结果都需要截图保存，在雨课堂提交为：20231000000+姓名+实验 2-1 创新.pdf）

- (1)菜单的选择模式，选择后不退出，直接提示，并等待下一次选择；
- (2)菜单选择后，下一次选择，记录上一次的选择菜单；
- (3)多级菜单显示，并支持返回到上一级菜单，也可直接返回主菜单；
- (4)其他的创新菜单模式。

```
COM4 - PuTTY

*****Choice Menu*****
[0] Open Device
[1] Left Shift
[2] Right Shift
[T] Stop Shift
[C] Close Device
[S] Sub Menu
[X] Exit Test
*****
your last choice: 1
Please input your choice: 2

The device is right!

*****Choice Menu*****
[0] Open Device
[1] Left Shift
[2] Right Shift
[T] Stop Shift
[C] Close Device
[S] Sub Menu
[X] Exit Test
*****
your last choice: 2
Please input your choice: s

----- Sub menu-----
[1] Speed Up
[2] Speed Down
[R] Return to Main Menu
please input sub menu number:r

---> return main menu--

*****Choice Menu*****
[0] Open Device
[1] Left Shift
[2] Right Shift
[T] Stop Shift
[C] Close Device
[S] Sub Menu
[X] Exit Test
*****
your last choice: s
Please input your choice: 
```

```
COM4 - PuTTY

.2.4:1.0/input/input5
[ 5.927183] generic-usb 0003:2A7A:4103.0001: input: USB HID v1.10 Keyboard [U
] on usb-s5p-ehci-3.2.4/input0
[ 5.953724] input: U as /devices/platform/s5p-ehci/usb1/1-3/1-3.2/1-3.2.4/1-3
.2.4:1.1/input/input6
[ 5.963735] generic-usb 0003:2A7A:4103.0002: input: USB HID v1.10 Device [U]
on usb-s5p-ehci-3.2.4/input1
[root@farsight]#
[root@farsight]#
[root@farsight]# ls
bin      etc      linuxrc  proc     sbin     tmp      var
dev      lib      mnt      root     sys      usr      www
[root@farsight]# ls
bin      etc      lib      mnt      root     sys      usr      www
dev      hello.c  linuxrc  proc     sbin     tmp      var
[root@farsight]# [ 63.522405] failed to copy MFC F/W during init
[ 63.711274] CPU1: shutdown

[root@farsight]#
[root@farsight]# ls
bin      hello.c      mnt      sbin      tmp      www
dev      lib          proc     sys      usr
etc      linuxrc      root     testmeun_arm  var
[root@farsight]# 
```

```
COM4 - PuTTY
bin      etc      linuxrc  proc     sbin     tmp      var
dev      lib      mnt      root     sys      usr      www
[root@farsight]# ls
bin      etc      lib      mnt      root     sys      usr      www
dev      hello.c  linuxrc  proc     sbin     tmp      var
[root@farsight]# [ 63.522405] failed to copy MFC F/W during init
[ 63.711274] CPU1: shutdown

[root@farsight]#
[root@farsight]# ls
bin      hello.c      mnt      sbin      tmp      www
dev      lib          proc     sys      usr
etc      linuxrc      root     testmeun_arm var
[root@farsight]# ./testmeun_arm
*****Choice Menu*****
[0] Open Device
[1] Left Shift
[2] Right Shift
[T] Stop Shift
[C] Close Device
[x] Exit Test
*****
Please input your choise: x
[root@farsight]#
```